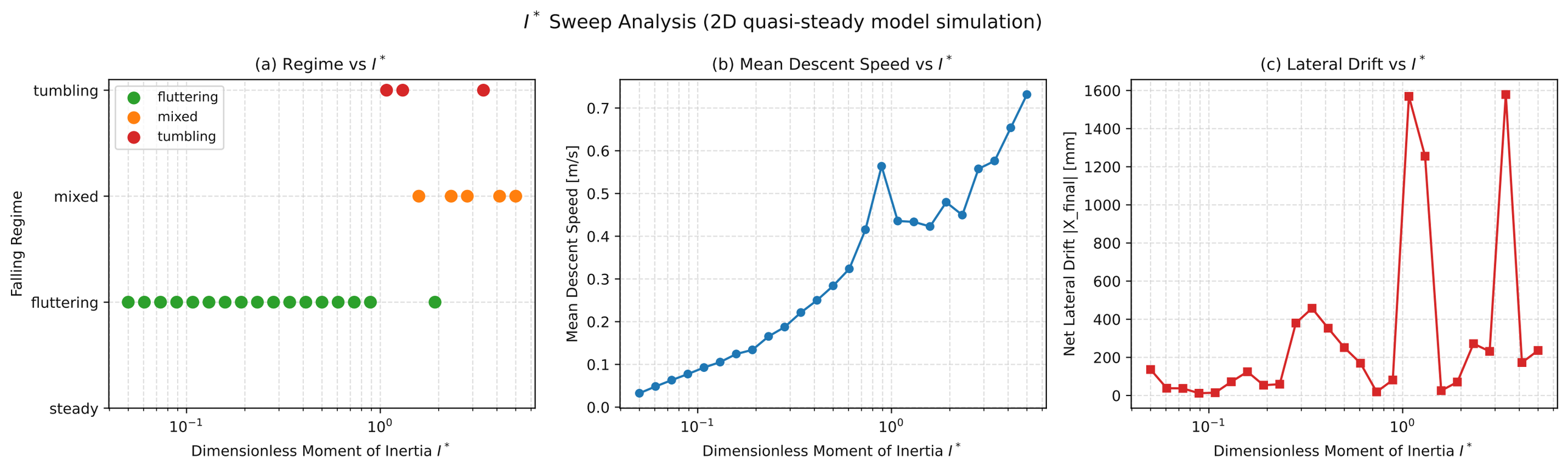
I^* Sweep Analysis (2D quasi-steady model simulation)
(a) Regime vs I^*
fluttering
mixed
tumbling
Falling Regime
steady
fluttering
mixed
tumbling
10^{-1}
10^{0}
Dimensionless Moment of Inertia I^*
(b) Mean Descent Speed vs I^*
Mean Descent Speed [m/s]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
10^{-1}
10^{0}
Dimensionless Moment of Inertia I^*
(c) Lateral Drift vs I^*
Net Lateral Drift |X_final| [mm]
0
200
400
600
800
1000
1200
1400
1600
10^{-1}
10^{0}
Dimensionless Moment of Inertia I^*